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A new passive PV heatsink design to reduce efficiency losses: A computational and experimental evaluation

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ABSTRACT

The efficiency of photovoltaic systems is decreased when its operating temperature increases, which can affect the performance in arid climatic conditions. In this work, a new passive cooling system for photovoltaic panels is presented, with the aim of reducing its operational temperature and preventing a reduction in its electrical efficiency in unfavorable environments. With this aim, a series of 3D models of heatsinks were designed. The systems were evaluated through simulating normal operational conditions using CFD software. Prototypes of the models with the best performance regarding level and distribution of temperature were built for experimental evaluation. The use of a segmented aluminum sheet was proposed; allowing a better airflow, and therefore, improved cooling under high irradiance conditions. Numerical simulation results showed improved in temperature distribution and a reduction of up to 9.4 °C in the most effective heatsinks. Experimental results were in close agreement, achieving a reduction of around 10 °C during peak irradiance. Results of this work support the use of the integration of PV and passive cooling systems to reduce efficiency losses under real operating conditions, where there is a multidirectional nature of airflow, which may not be favorable for a conventional heatsink with a continuous channel design.

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1. Introduction

Photovoltaic systems (PV) are widely used for power generation; however, most commercial solar modules currently have a low conversion efficiency, and only convert between 13 and 20% of incident solar radiation to electricity under ideal operational conditions [1,2]. The remaining component of radiation, absorbed by the cells, is transformed into thermal energy, causing an increase in its temperature. Temperature plays an important role in PV system efficiency, as solar module performance is directly related to the operating temperature of its PV cells. An increase in temperature produces a reduction in both the open circuit voltage and fill factor, as well as an increase in the recombination of internal charge carriers, resulting in a decrease in the output power of the PV system [3–7]. The increase in temperature affects energy production in the short term; however, it has also been found that thermal cycles accelerate the aging of the equipment in the long term and degrade its performance [8,9].

The amount of energy that a PV module can produce depends mainly on its type, composition, and environmental factors (mainly temperature and solar irradiance). The modules are tested and certified under laboratory conditions to establish their nominal performance (Standard Test Conditions, AM = 1.5, irradiance of 1000 W/m^2 and module temperature of 25 °C). However, these conditions are rarely met in a real environment, and PV modules generally do not reach their nominal performance [10]. Additionally, exposure of PV systems to environmental elements degrades their performance. It has been documented that temperature is related to several types of degradation such as: delamination, hot spots, corrosion, glass fracture, electrical migration between module layers/interconnections and discoloration [11].

Crystalline silicon PV cell modules are the most widely used worldwide, due to the cost and proven reliability. However, the

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Nomenclature

ρ	density (kg/m^3)
$\vec{v}$	velocity (m/s)
$\vec{g}$	gravitational acceleration (m/s^2)
E	energy (J)
t	time (s)
p	pressure (Pa)
$\vec{q}$	heat flux (W/m^2)
k	thermal conductivity (W/m K)
T	temperature (K)
q_{conv}	natural convection heat loss (W/m^2)
h_{conv}	heat transfer coefficient (W/m^2 K)
T_s	heated surface temperature (°C)
T_{amb}	ambient temperature (°C)
q_{rad}	radiative heat loss (W/m^2)
ε	emissivity of glass surface
σ	Stefan-Boltzmann constant ($\sigma = 5.67 \times 10^{-8}$ W/m^2 K^4)
T_0	surrounding absolute temperature (°C)

efficiency of this technology depends on its operating temperature, especially in hot environments where, for every 1 °C increase in the solar module, a loss of electrical power of 0.5% could affect the system [11–16].

An effective way to counteract the rate of thermal degradation of a PV module is by reducing its surface temperature. This can be achieved by cooling the system (or preventing it from heating up) during operation [17,18]. According to published studies, for their application in PV modules, cooling systems can be classified as active or passive. Previous work has shown that active systems can be very effective [19–25]; however, its use results in increased cost due to its complexity and need for maintenance, in addition to consuming energy by relying on mechanical/electrical systems, such as pumps, fans and sensors to maintain a working fluid in circulation. Rekha et al. [23] used Computational Fluid Dynamics (CFD) software to simulate and evaluate the performance of a forced convection (air) photovoltaic/thermal (PV/T) hybrid system. Their results were validated with experimental works reported in the literature, and good agreement was found between the numerical and experimental results for the air temperature in the system and the temperature of the PV cells. Using the validated model, the effect of the mass air flow and the channel depth on the performance of the hybrid system PV/T were studied. The proposed design showed a 20% performance improvement when compared to a conventional PV/T air collector. Mojumder et al. [24] proposed a hybrid PV/T collector that integrates rectangular fins as heat dissipating elements. The average temperatures of the surfaces of the PV module, the collector and the outlets and inlets were measured experimentally, varying the number of fins (0–4), mass flows (0.02–0.14 kg/s) and radiation levels (200–700 W/m^2). The maximum thermal and photovoltaic efficiencies obtained were 56.19% and 13.75%, respectively, using four fins with a flow of 0.14 kg/s and 700 W/m^2 of solar radiation.

Passive cooling systems depend on three basic mechanisms of heat transfer: convection, conduction, and radiation, which operate without consuming energy when compared to active systems [26]. According to previous studies, natural air convection is the simplest method to cool a PV module; the surrounding air acts as a working fluid and absorbs the heat of the equipment circulating naturally. Other methods include the thermosyphon effect (liquids) and, recently, the use of Phase Change Materials (PCMs) [27–31]. Popovici et al. [29] carried out a numerical study of a channeled heatsink of copper fins varying the length and inclination. For the different study cases, the fin height was modified from 0.01 m to 0.05 m in increments of 0.01 m. Additionally, perforations were added to the fins to facilitate air flow. The inclination of the fins was also varied with respect to the horizontal, using inclinations of 45°, 90° and 135°. The remaining geometric parameters of the heatsink remained fixed. For climatic conditions, an ambient temperature of 35 °C, irradiation of 500 W/m^2, and an air speed of 1.5 m/s perpendicular to the channels was used to generate turbulence. The results indicate that the panel without heatsink reaches a maximum temperature of 56 °C, with an electrical efficiency of 86% of its rated power. When the heatsink was added, the temperature was reduced by around 10 °C, which allowed an electrical performance greater than 90% of the rated power to be achieved. The most effective fins were those inclined at 45 °C, increasing the maximum power by 7.55% when compared to the reference. Chandrasekar and Senthilkumar [30] tested a passive system that used a heat spreader in conjunction with cotton wick. The maximum temperature of the module was reduced by 5.9 °C (from 49.2 °C to 43.3 °C). This was achieved with the combined effect of evaporative cooling and the heat diffuser fins mounted on the back of the PV module. The reduction of operational temperature resulted in an improvement of 14% in electrical power.

Table 1 presents a qualitative comparison of the different cooling systems found in the literature.

It was found that many systems are effective to reduce and maintain a uniform temperature level of the PV system. However, due to cost and complexity, they are limited to solar concentrated photovoltaic (CPV) applications. Given that, the aim of this work is to design a simple, low cost cooling system for use in commercial PV modules, proposing a new passive heatsink. Commonly, heatsinks are manufactured from solid blocks of aluminum, and are used to cool electronic devices; this entails certain disadvantages for use in solar PV modules. Coupling a heatsink made with conventional techniques to a commercial PV module (250–300 W) would be very expensive, in addition to considerably increase the weight of the system. Moreover, conventional heatsink geometry (continuous profile fins) is a design intended for specific operating conditions, such as forced convection with constant working fluid speed and direction.

The present work proposes a new fin array for PV passive heatsinks, which increases vortex generation and reduces pressure losses across heatsink channels by allowing a less restrictive airflow; enhancing the convective heat exchange and cooling down the PV system. The use of multiangular aluminum fins (segmented fin profile) is evaluated as a passive cooling system that reaches a relatively homogeneous temperature distribution in a PV system when compared to a conventional finned heatsink (continuous fin profile), by maintaining a more stable thermal response under different directions of wind flow incidence. This can have an impact on the lifetime and efficiency of the PV cell. The optimization of material and heat transfer was studied by evaluating the length of the fin with a wind attack in the three main directions at 0°, 90° and 45°, assessing the cost/benefit relationship. A numerical model was used in ANSYS FLUENT to study the effect of the variation of design parameters and environmental variables in the reduction and uniformity of temperature. The findings of the study are relevant to this field because they demonstrate the potential for making PV systems work more efficiently, especially in hot climates.

2. Proposed PV cooling system

The proposed model with fins interleaved at ±30° with respect to the vertical axis (see Fig. 1) uses a commercial aluminum (1100

Table 1
Main qualities of different cooling systems in literature [32–35].

Cooling system	Advantages	Disadvantages
Photovoltaic Thermal (PV/T)	• Increased combined efficiency. • Extracted heat can be used for domestic applications. • Increased energy yield.	• High cost. • Requires mechanical/electrical systems to circulate the working fluid when passive cooling techniques are not used (i.e. thermosyphon effect). • Needs regular maintenance.
Heatsinks	• Little to no maintenance needed. • Mostly used for passive cooling (has no moving parts nor consumes energy). • Can achieve good temperature uniformity.	• Extracted heat is wasted. • Conventional fabrication techniques (extrusion/CNC machining) limits its use to mainly concentrated photovoltaics (CPV). • Its performance depends on the environmental conditions.
Phase Change Material (PCM)	• Usually used in conjunction with another cooling system. • Can store large amounts of heat. • Aids to achieve good temperature uniformity.	• Heat absorption capabilities degrades over time. • Some materials have a lower thermal conductivity. • Less efficient in cold climatic conditions.
Jet impingement	• Can extract a large amount of heat. • Can be used in conjunction with another cooling system. • Proven efficiency, given its widely used in several industrial fields.	• High cost. • Complex to design, due to the number of parameters that must be optimized. • Difficult integration with PV modules.
Heat pipes	• Can transfer heat with minimal losses. • Passive cooling device. • Can achieve good temperature uniformity.	• High cost. • Mainly limited to CPV systems due to scalability. • Difficult integration with PV modules.
Liquid immersion	• Economic. • Can reduce optical losses. • Can maintain low and uniform PV cell temperature.	• Conversion efficiency depends on depth of immersion, velocity and temperature of the medium. • Low efficiency during cloudy days. • Due to electrical conductivity, ionized liquids cause efficiency losses.
Micro channels	• Can extract a large amount of heat. • Can be used in conjunction with another cooling system. • Can achieve good temperature uniformity.	• High cost. • Mainly limited to CPV systems due to scalability. • Requires mechanical/electrical systems to circulate the working fluid to a heat exchanger.

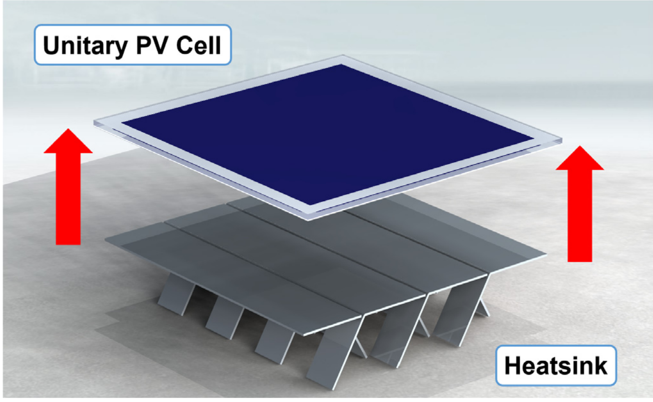

Fig. 1. PV cooling system representation.

alloy) of 20 gauge, which is die cut and mounted on an aluminum absorber plate for evaluation and comparison with its conventional profile counterpart (continuous at 90°). The heatsink geometry is designed to maintain consistent performance in situations where the flow and direction of the wind varies continuously. Part of the study contemplates the thermal response in different scenarios; without heatsink, with conventional heatsink and multi-angular fin heatsink at ±30° with respect to the vertical axis (60° between fins).

3. Mathematical model

3.1. Governing equations

The heat transfer in an air-cooled PV module involves multiple solid domains. The fluid domain includes the working fluid inside the heatsink (air). There is a solid domain for each layer of the module material (glass, cell, EVA, Tedlar) and for the heatsink. The physical aspects of the fluids are governed by the following principles: conservation of the mass, conservation of the moment and conservation of energy. These principles can be expressed as a set of mathematical expressions called Navier-Stokes equations:

Continuity:

$$\frac{\partial \rho}{\partial t} + \nabla\cdot(\rho \vec{v}) = 0 \tag{1}$$

Moment:

$$\frac{\partial}{\partial t}(\rho \vec{v}) = -\nabla p + \rho \vec{g} \tag{2}$$

Energy:

$$\frac{\partial}{\partial t}(\rho E) + \nabla\cdot(\vec{v}(\rho E + p)) = -\nabla\cdot\left(\sum_j h_j J_j\right) \tag{3}$$

where ρ is density, $\vec{v}$ velocity, and E denotes energy. The governing equations are discretized using the standard method of finite volumes to form a system of algebraic equations which is solved by an iterative process using Computer Fluid Dynamics software.

3.2. Thermal load

The thermal model predicts the temperature field through the thickness of the unit cell. Given the reduced thickness of the layers compared to their length and width, the thermal exchanges towards the edges of these are neglected. Conductive heat losses are caused by gradients between the cell and other materials with which it is in contact (including the surrounding air). The ability of the device to transfer heat to the environment is described by its thermal resistance and the configuration of the materials used to encapsulate the PV cell. If there is a temperature field in a medium $T(x, y, z, t) = T(\vec{r}, t)$, Fourier's law expresses instantaneous heat fluxes as:

$$\vec{q} = -k\nabla T \tag{4}$$

where k is the material thermal conductivity and $\vec{q}$ the heat flux.

This heat transfer mechanism allows the modeling of energy transmission between the layers of different materials and thicknesses of the unit cell.

Since the proposed thermal model considers different environmental conditions (solar radiation, ambient temperature, wind speed, etc.), heat exchanges can be described in terms of two physical phenomena, convection and radiation. Convection by air is expressed as:

$$q_{conv} = h_{conv}(T_s - T_{amb}) \tag{5}$$

and the radiative exchanges with the sky or ground, as:

$$q_{rad} = \varepsilon \times \sigma\left(T_s^4 - T_0^4\right). \tag{6}$$

In Eq. (5), h_{conv} is the heat transfer coefficient at surface temperature T_s. This convection is produced by the air at ambient temperature T_{amb}. It should be noted that h_{conv} could be expressed differently for natural or forced convection. In Eq. (6), ε is the glass surface emissivity, σ is the Stefan-Boltzmann constant, and T_0 the surrounding absolute temperature.

3.3. Methodology

The computational methodology used (Fig. 2) for obtaining and evaluating an optimized heatsink consists of three steps:

(I) **CFD model resolution.** A schematic of the problem is presented in Fig. 2, in which the elementary parts are delimited, i.e. the physical phenomena that interact with the PV system. This problem was converted into a computational domain, in which, a unit cell of 170 × 170 mm^2 was modeled. The model contains the same standard layers of a conventional PV module and considerably reduces the number of elements in the mesh, which significantly accelerates the resolution of the models. The properties of the modeled layers are detailed in the Table 2. Hybrid meshes were used, given the complex geometry of the volume, that includes the fins and the air mass in the models with a heatsink, which could only be adequately meshed with tetrahedral elements; the rest of the layers were made with hexahedral elements. A mesh dependence study was conducted to find the appropriate element size. With the necessary mesh refinement, the number of elements went from 1.15 × 10^6 for the unit cell without a heatsink to 2.94 × 10^6 for the cell with the most complex heatsink.

The boundary conditions and parameters of the model are detailed in the following Table 3:

To reduce mesh elements and lighten the modeling processes, a correlation which gave a good time/precision compromise was used to calculate the convective coefficient in the upper surface of the unit cell [37]:

$$h = 5.7 + 3.8v \tag{7}$$

Before the numerical simulations, the following assumptions were made:

- Ambient temperature is constant and equal on all sides of the unit cell and was calculated based on a history of three years and corresponds exclusively to the average for hours of sunlight.

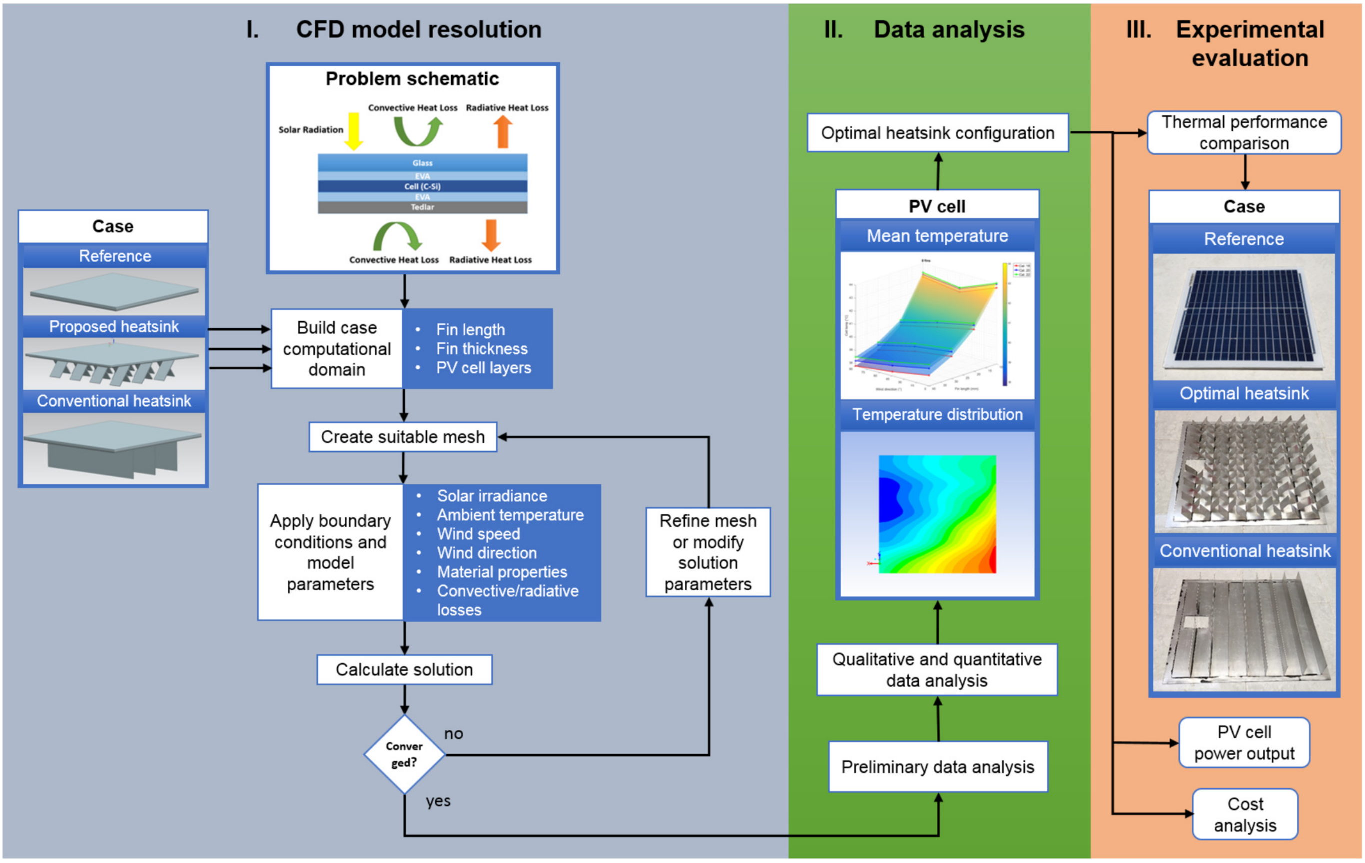
I. CFD model resolution
II. Data analysis
III. Experimental evaluation
Problem schematic
Solar Radiation
Convective Heat Loss
Radiative Heat Loss
Glass
EVA
Cell (C-Si)
EVA
Tedlar
Convective Heat Loss
Radiative Heat Loss
Case
Reference
Proposed heatsink
Conventional heatsink
Build case computational domain
Fin length
Fin thickness
PV cell layers
Create suitable mesh
Apply boundary conditions and model parameters
Solar irradiance
Ambient temperature
Wind speed
Wind direction
Material properties
Convective/radiative losses
Refine mesh or modify solution parameters
Calculate solution
Converged?
no
yes
Optimal heatsink configuration
PV cell
Mean temperature
Temperature distribution
Qualitative and quantitative data analysis
Preliminary data analysis
Thermal performance comparison
Case
Reference
Optimal heatsink
Conventional heatsink
PV cell power output
Cost analysis

Fig. 2. Computational methodology of the heatsink modeling process and data analysis to obtain an optimal configuration for performance evaluation.

Table 2
Thermo-physical properties of the unitary cell/PV module layers [36].

Layer	Thickness (mm)	**Thermal conductivity (W/m K)**	Density (kg/m^3)	Specific heat (J/kg°C)
Glass	3	1.8	3000	500
Cell	0.3	148	2330	677
EVA	0.5	0.35	960	2090
Tedlar	0.1	0.2	1200	1250

Table 3
Boundary conditions, PV cell characteristics and heatsink parameters.

Environmental		
Boundary Condition	Value	Unit
Solar irradiance	1000	W/m^2
Ambient temperature	28	°C
Wind speed	2	m/s
Wind direction	0°, 45°, 90°	
PV cell		
Boundary Condition	**Value**	**Unit**
Front surface emissivity [36]	0.91	
Front surface convective coefficient	13.3	W/m^2 K
Heatsink		
Geometric Parameter	**Value**	**Unit**
Fin length	10, 20, 30, 40	mm
Fin width	20	mm
Fin spacing	42	mm
Sheet gauge (fin thickness)	22 (0.71)	(mm)
Fin angle	±30° with Y axis	

- Wind speed is constant at the model inlet and any changes in speed within the model fluid domain and outlet is due to the interaction with the unit cell/heatsink geometry. The value of 2 m/s was deemed appropriate given that this is a typical value of the studied region.
- No dirt, that affects the absorptivity of the cell, is considered.
- All materials in the unit cell have isotropic properties.
- The problem is considered in 3D and steady state.
- The upper surface of the cell is exposed to a constant and homogeneous flow of solar radiation.

(II) **Data analysis.** Qualitative results are given in the form of temperature distribution. This is used to observe the homogeneity in the temperature distribution of the PV cell, as well as the difference between maximum and minimum temperatures. The quantitative data corresponds to average, maximum and minimum temperatures on the surface of the PV cell calculated by the computational model. This allows for more accurate calculations of temperature reduction and power delivery of the PV cell when compared to a visual inspection of a temperature contour plot. Furthermore, this data allows the identification of the parameters of the heatsink, including the optimal configuration to achieve a greater temperature reduction, as well as improved homogeneity in its distribution.

(III) **Experimental evaluation.** Two prototypes were built; one of the heatsinks was considered optimal, while the other contained a conventional geometry counterpart (continuous fin profile), elaborated with the proposed manufacturing technique. The systems are coupled to a solar PV module, to be evaluated under field conditions in terms of its temperature reduction and distribution homogeneity. The power obtained using the heatsink is compared with the power of the reference PV panel. Additionally, the costs are calculated based on the use of material for the given array.

4. Simulation work

4.1. Simulation results

A summary can be observed in Table 4 that shows four heatsink variants and the reference PV cell, each one being evaluated from three airflow directions (0°, 45° and 90°). The temperature in the cell varies according to the direction of the airflow; the lowest temperature observed corresponds to the best case, and the highest to the worst case. Generally, the temperature is lower when the air flows at 90° (perpendicular to the fins), and higher when the air flows at 0° (parallel to the fins). The maximum temperature reduction obtained is around 9.4 °C (19.6%), which corresponds to the heatsink with 40 mm fins.

It should be noted that the increase in fin length of the heatsink does not correspond linearly with the reduction of temperature. The thermal efficiency of a fin decreases exponentially when increasing its length, as less heat is transmitted to the environment when moving away from the base of the fin. Consequently, with each increment, the reduction in temperature will become increasingly smaller, thus extra use of material may not be justified considering costs and practicality. Due to this, it can be observed in Table 4 that the reduction of temperature decreases with each length increase, and in order to double the performance of the 10 mm fin it is necessary to extend more than four times its length.

4.2. Thermal effect of fin length

Fig. 3 shows the thermal distribution of the PV unitary cell, with and without fins (top view) of the studied element (170 × 170 mm^2) used as a constant for this study; the airflow direction is at 45°, with 20 mm fin width and 0.71 mm thickness, varying only the fin length (the fins are not observed in this view). The probe tool was used during post-processing to get an accurate measurement of temperature differences between higher and lower temperature regions of the PV cell. In Fig. 3a the temperature distribution of the unitary cell without fins has a gradient from side to side of 9 °C. In Fig. 3b the gradient using 10 mm fins is reduced to 4.9 °C, in Fig. 3c (20 mm) to 4 °C, in Fig. 3d (30 mm) to 3.8 °C, and in Fig. 3e (40 mm) to 3.6 °C. In qualitative terms, as the length of the fins increases, the region of lower temperature tends to increase (blue tones). It is also evident that, to some extent, increasing the length of the fins no longer produces significant changes in temperature reduction where the most obvious change is observed, i.e. when increasing from 10 mm to 20 mm (Fig. 3b y Fig. 3c). On the other hand, when increasing from 30 mm to 40 mm (Fig. 3d y Fig. 3e), the change in temperature distribution is minimal as well as the reduction in its gradient (0.2 °C).

It is observed that, when adding the heatsink, a higher temperature homogeneity of the PV cell is obtained, and this effect decreases significantly by extending the length of the fins beyond 30 mm. Fins with 40 mm length achieve the best temperature distribution of the PV cell with the lowest thermal gradients. Also, given the spacing between fins (42 mm) and fin angle (±30° with respect to the vertical axis), increasing the length beyond 40 mm would produce a high geometrical density, thus, trapping the heat.

4.3. Effect of airflow direction

Fig. 4 illustrates the thermal response of the proposed heatsink, modeled with a 20 mm fin width (fin thickness of 0.71 mm), as well

Table 4
Summary of theoretical results (mean PV cell temperatures).

Heatsink fin length (mm)	Cell Temperature (°C)		Temperature Reduction (°C)		Temperature Reduction (%)	
	Best case	Worst case	Best case	Worst case	Best case	Worst case
Reference (no heatsink)	47.92	48.65	0	0	0	0
10	42.91	44.02	5.01	4.63	10.45	9.51
20	40.21	41.49	7.71	7.16	16.09	14.71
30	39.06	40.10	8.86	8.55	18.49	17.57
40	38.49	39.42	9.43	9.23	19.67	18.97

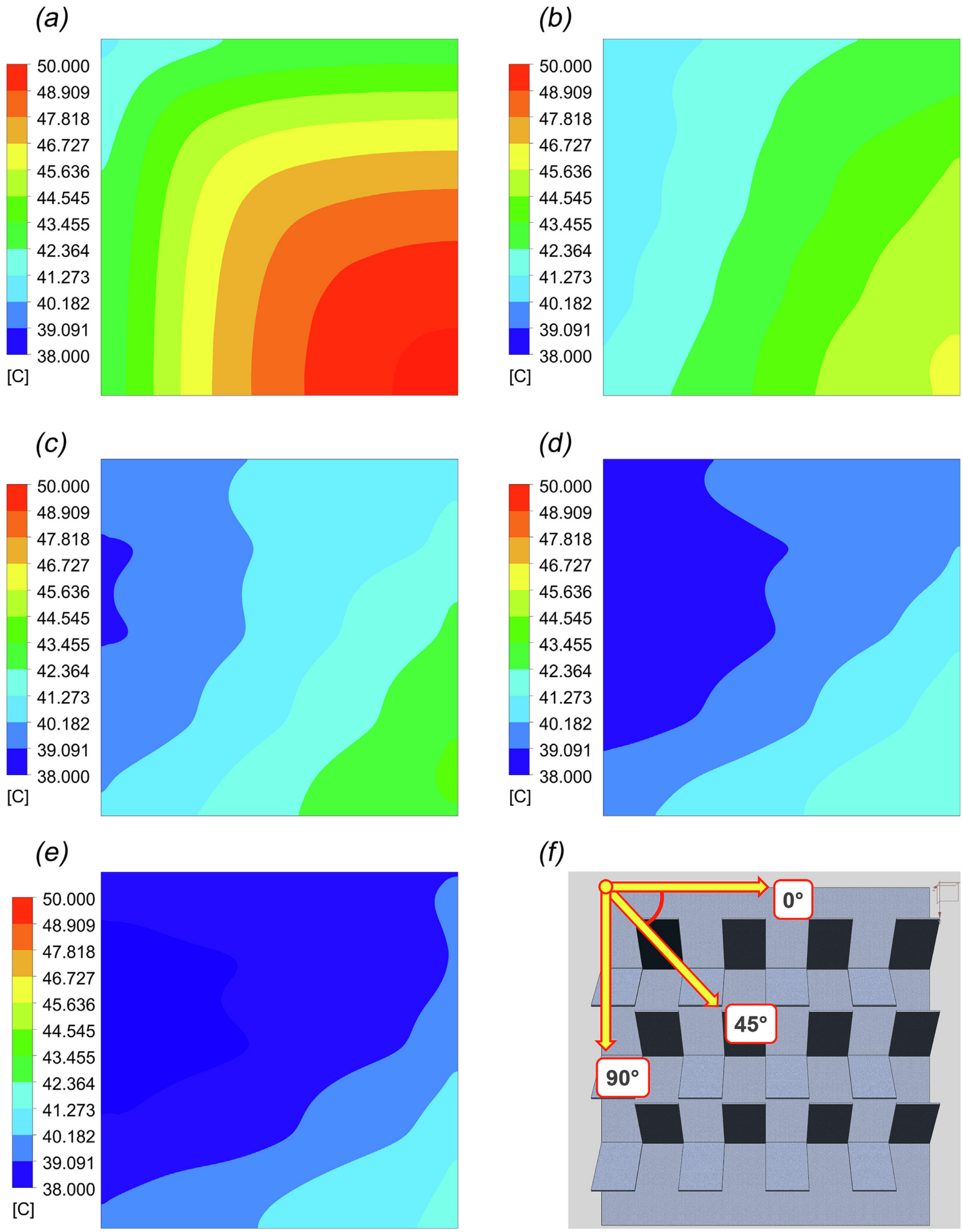

Fig. 3. Simulation of temperature distribution of the PV unitary cell using different fin lengths, (a) No heatsink, (b) 10 mm fin, (c) 20 mm fin, (d) 30 mm fin, (e) 40 mm fin, (f) Fin orientation and airflow direction.

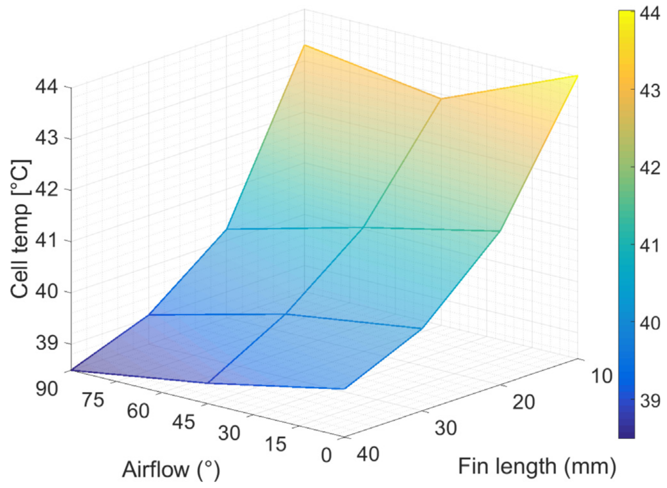

Fig. 4. Simulation response of PV cell temperature by influencing airflow direction and fin length.

as fin lengths of 10, 20, 30 and 40 mm. The relation between airflow direction, cell temperature and fin length are studied. It is observed that the proposed heatsink has a favorable behavior when varying the direction of airflow; a reduction of the temperature is observed when the air flow changed direction from 0° to 90°.

Generally, the temperature of the cell is lower when the airflow is at 90° (perpendicular to the face of the fins), except when the length of the fins is 10 mm, where the thermal response resembles that of the reference PV cell (the temperature is lower with an airflow at 45°). In the reference PV cell, the diagonal is the greatest distance that the air can travel in contact with its surface, favoring a better heat transfer and, therefore, a greater heat extraction. The use of fins with a length of 10 mm does not alter the airflow greatly, thus, this behavior is maintained (as without fins). This can be seen in Fig. 3a and Fig.3b, where the temperature contours for the reference cell and the 10 mm finned heatsink are similar. On the other hand, by elongating the fins, the airflow is altered greatly, affecting the velocity, turbulence and air pressure behavior inside the heatsink channels, causing a more stable thermal response, particularly for the fins with a length of 40 mm. These aspects are discussed in more detail in the next section.

4.4. *Comparison of proposed heatsink vs conventional heatsink*

The aim of the present work is to obtain a heatsink that is more efficient than a conventional passive heatsink with straight fins (L shape). For comparative purposes, a heatsink with conventional profile fins was modeled, considering the same proposed manufacturing technique (sheet stamping). Both designs are based on the 22 gauge aluminum sheet (0.71 mm) and fins with 40 mm length, as these were the most cost effective. Both heatsinks use the same amount of material. The difference is that the proposed model is segmented with opposite fins with an inclination of ±30° with respect to the vertical axis, allowing, in the best-case scenario, a more uniform air flow. For this to be observed, three representative incidence directions of air flow were evaluated (Fig. 5).

In Fig. 6a the proposed model performs better than a conventional heatsink, especially when the airflow of 2 m/s approaches 0° in direction (parallel to the fins). For each direction of the flow studied (0°, 45° and 90°), the resulting temperatures are 39.13 °C, 38.59 °C and 38.18 °C, respectively, for the proposed heatsink, and 40.12 °C, 38.82 °C and 38.74 °C, respectively, for the conventional heatsink.

When the air flows parallel to the fins (0°), the pressure loss is minimum for both heatsinks (Fig. 6b); however, the segmentation of the fins allows for an increase in the area of heat dissipation. In addition, this increases slightly the level of turbulence in the heatsink channels without significant reduction in the airflow velocity, resulting in a greater temperature reduction when compared to a conventional geometry. When the incidence of the air is diagonal (45°), both heatsinks perform similarly, given the fins tend to redirect the airflow. However, the segmented fin heatsink allows a less restrictive airflow, resulting in lower pressure losses and improved heat transfer. The most significant case occurs when the air flows perpendicular to the fin channels (90°). This scenario merits a more detailed analysis given it is the worst-case scenario for both heatsinks, due to high-pressure losses, particularly for the conventional heatsink.

A working fluid loses pressure when flowing through a heat exchanger. If the losses are large, it may be due to a non-optimal design of the heatsink, which produces a low uniformity in the airflow, resulting in poor heat transfer. In Fig. 7a a pressure contour comparison at a 90° airflow direction is presented (heatsink cross section view). The proposed heatsink has a lower and less abrupt pressure loss, owning to a less restrictive design that allows a better

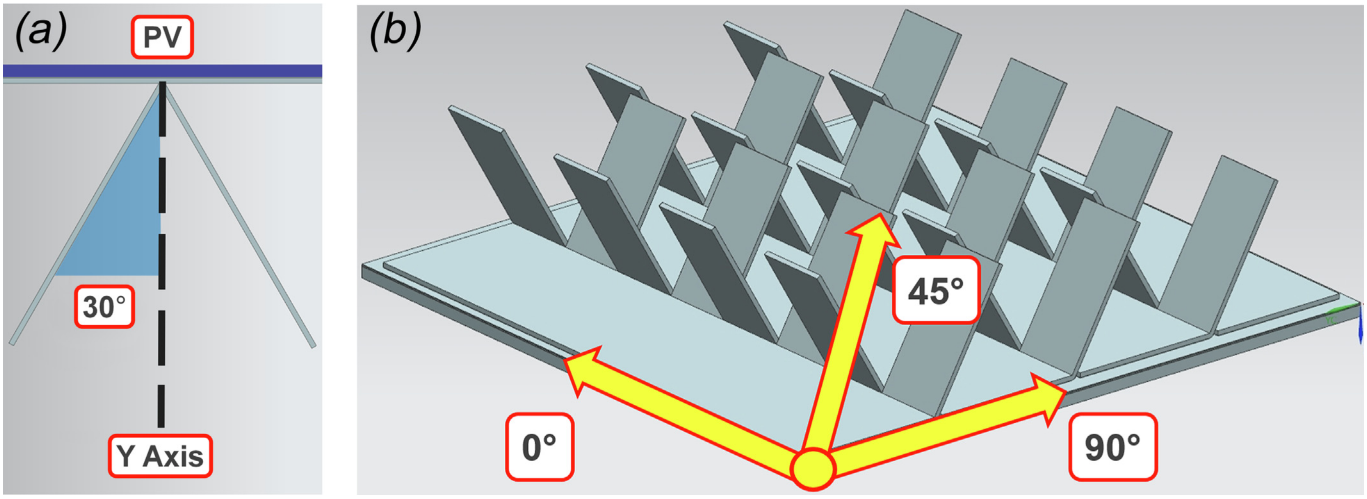

Fig. 5. (a) Fin angle used, (b) Airflow directions of study.

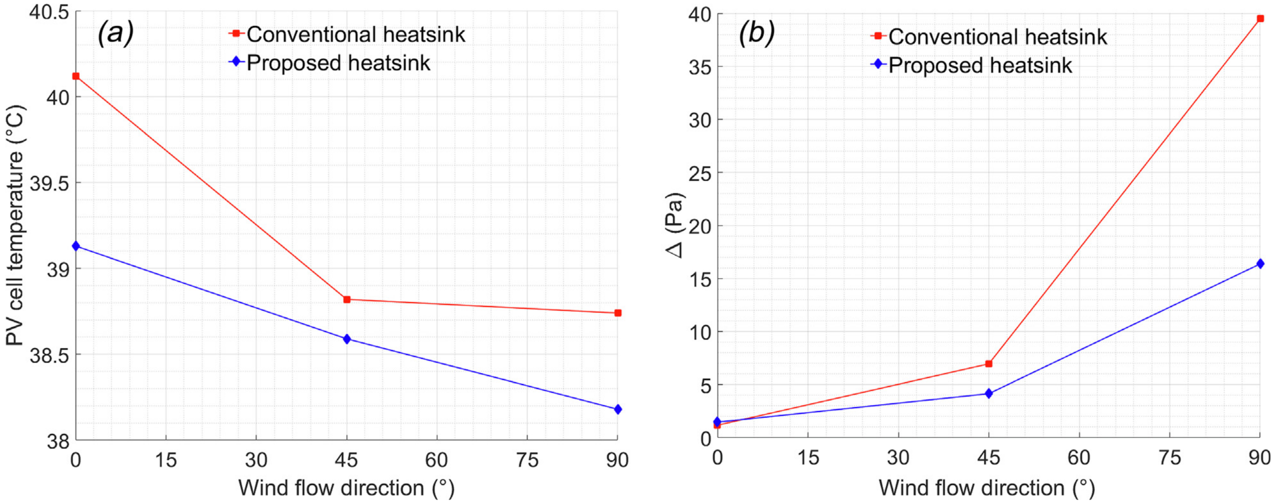

Fig. 6. Heatsink comparison for airflow at 0°, 45° and 90°, (a) PV cell mean temperature, (b) Heatsink pressure drop.

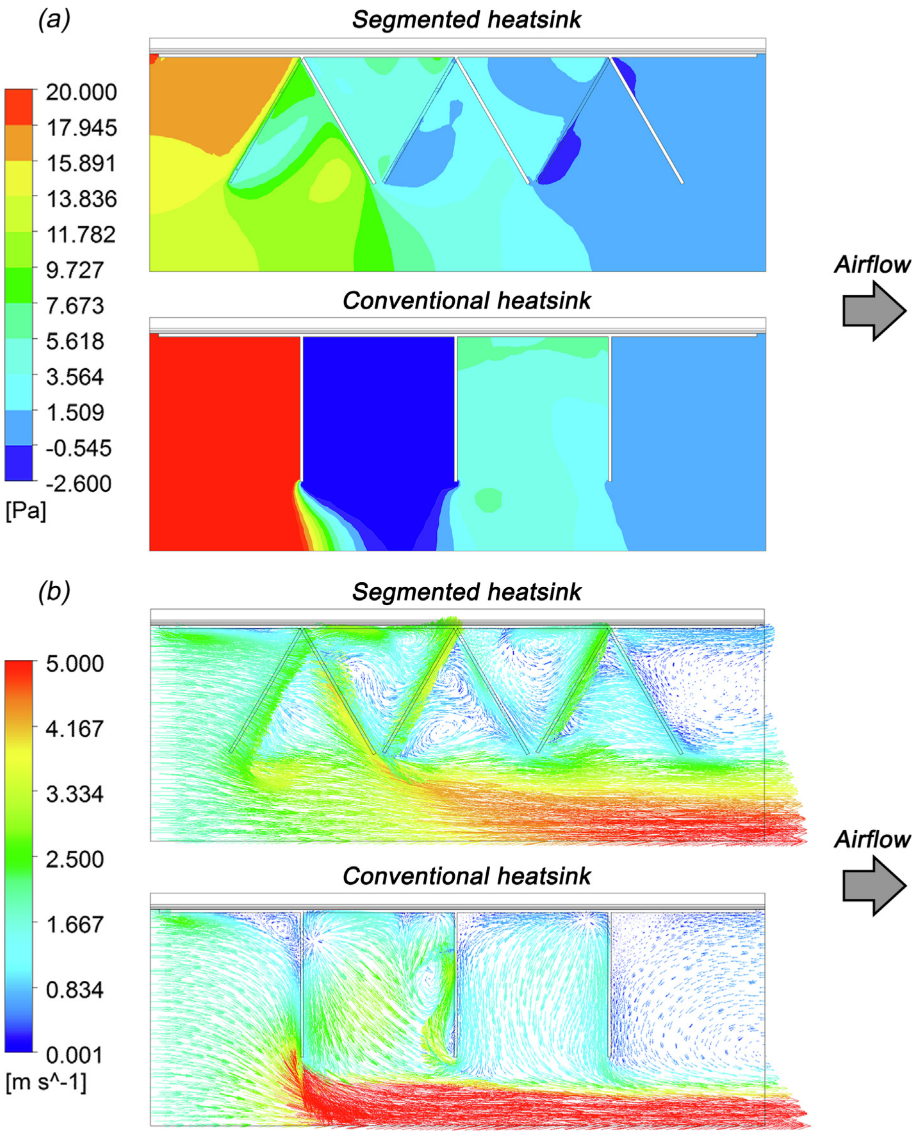
(a)
Segmented heatsink
Conventional heatsink
Airflow
20.000
17.945
15.891
13.836
11.782
9.727
7.673
5.618
3.564
1.509
-0.545
-2.600
[Pa]
(b)
Segmented heatsink
Conventional heatsink
Airflow
5.000
4.167
3.334
2.500
1.667
0.834
0.001
[m s^-1]

Fig. 7. Heatsink flow field with airflow at 90° (perpendicular to fin channels), (a) Heatsink pressure distribution, (b) Heatsink velocity distribution.

interaction of the fluid with the dissipation surface. This characteristic is significant, as, when scaling the system in a larger PV array, the loss of pressure will tend to increase with the size of the heatsink.

In Fig. 7b the velocity distribution is observed around the heatsinks. The proposed system shows a greater presence of swirls inside the heatsink, as well as a more gradual decrease in air velocity (more uniform flow). On the other hand, the conventional geometry produces an acceleration of the fluid, which is directed primarily out of the heatsink channels. This is unfavorable given it produces areas of low/negative pressure (Fig. 7a) and low speed immediately after the first fin, due to a vortex choking the airflow in the inlet region and altering greatly the speed distribution along the fin channels. In general, it is observed that the thermal performance of the heatsinks has a high correlation with its hydraulic performance.

As the airflow changes direction from parallel to perpendicular to fin channels, there is an increase in Reynolds and Nusselt numbers due to a more turbulent flow, thus, enhancing convective heat transfer despite of a higher pressure drop (Fig. 6).

5. Experimental and economic heatsink evaluation

5.1. Experimental setup

A preliminary experimental validation of the study is shown in Fig. 8. Two 15 W Dokio DSP-15P polycrystalline silicon solar panels (35 × 36 cm^2) were used to record its temperature when exposed to natural weather; three type K thermocouples were attached to the rear side (on each panel) in a diagonal arrangement. The panels were mounted on a custom-made base at an inclination of 21° (corresponding to the local latitude), and they were oriented towards the south. An ADAM data acquisition module model 6018 was programmed to sample the thermocouples at a frequency of 1 Hz. Environmental data such as temperature, irradiation, direction and air speed were obtained from a weather station located at the same site. The data was stored for further processing on a desktop computer.

Two 1100 aluminum alloy heatsinks were fabricated from 22-gauge (0.71 mm) aluminum sheet commercially available. Both heatsinks have a 40 mm fin length and use the same amount of material. The conventional variant has straight fins with a continuous profile and 32 cm width, the proposed model has fins with a segmented profile every 2 cm (16 fins per profile) with an interleaved inclination of ±30° with respect to the vertical axis. For convenience, during the testing stage the fins were attached to a flat plate of the same material (Fig. 8b) using epoxy resin with metal powder (Devcon R2-42). This arrangement was then attached to the PV panel using thermal paste, making its removal from the panel possible for use in further studies. This allowed for the testing of various heatsink arrangements in a more practical and cost-effective way without the need to irreversibly modify a PV panel.

The experiment was carried out on an obstacle-free rooftop in the city of Merida, Mexico. The measurements were taken in February from day 21–28, and in April from day 1–7 from 6:00 a.m. to 6:00 p.m., according to the availability of sunlight.

5.2. Experimental results

Measurements were carried out over 12 h, first comparing the PV panels under normal conditions as a reference point. Following this, the heatsinks were added, comparing each prototype individually against a reference panel, thus comparing both heatsinks. For the test days, the ambient temperature range was 22–34 °C, the peak solar irradiance was 810–970 W/m^2, and the wind showed a most probable speed of 4.5 m/s according to the distribution of Weibull with a predominantly south, south-east direction. However, it was observed that during peak irradiance hours (12:00–14:00), the wind speed dropped consistently, and for that period of time, the most probable speed was 2.7 m/s.

In Fig. 9a the irradiance corresponding to the segmented heatsink test vs the reference panel is observed (Fig. 9b). It is noteworthy that the irradiance directly affects the instantaneous temperature. The effect of the heatsink becomes more noticeable as the irradiance increases; that is, the whole system tends to acquire thermal equilibrium at dawn and dusk, with the effectiveness of the heatsink being most noticeable in the solar peak, reducing the temperature of the PV panel in these conditions, by around 10 °C (18%) with respect to the reference.

It was also observed that, during interruptions due to clouds, the temperature of the reference panel decreased to resemble that of the cooled system, that also decreased, but to a lesser extent.

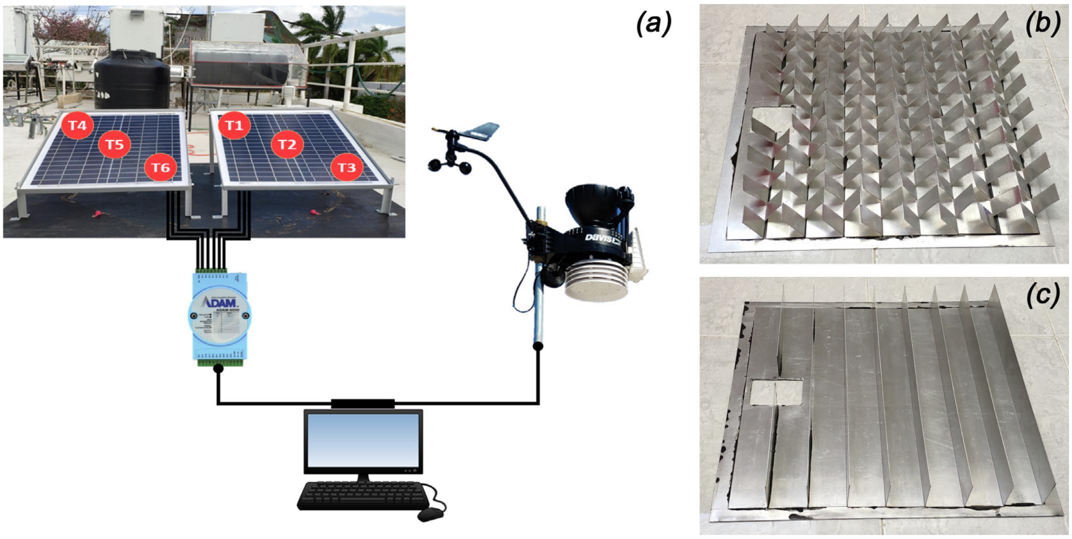

Fig. 8. Experimental setup for fin comparison, (a) Setup schematics, (b) Segmented heatsink, (c) Conventional heatsink.

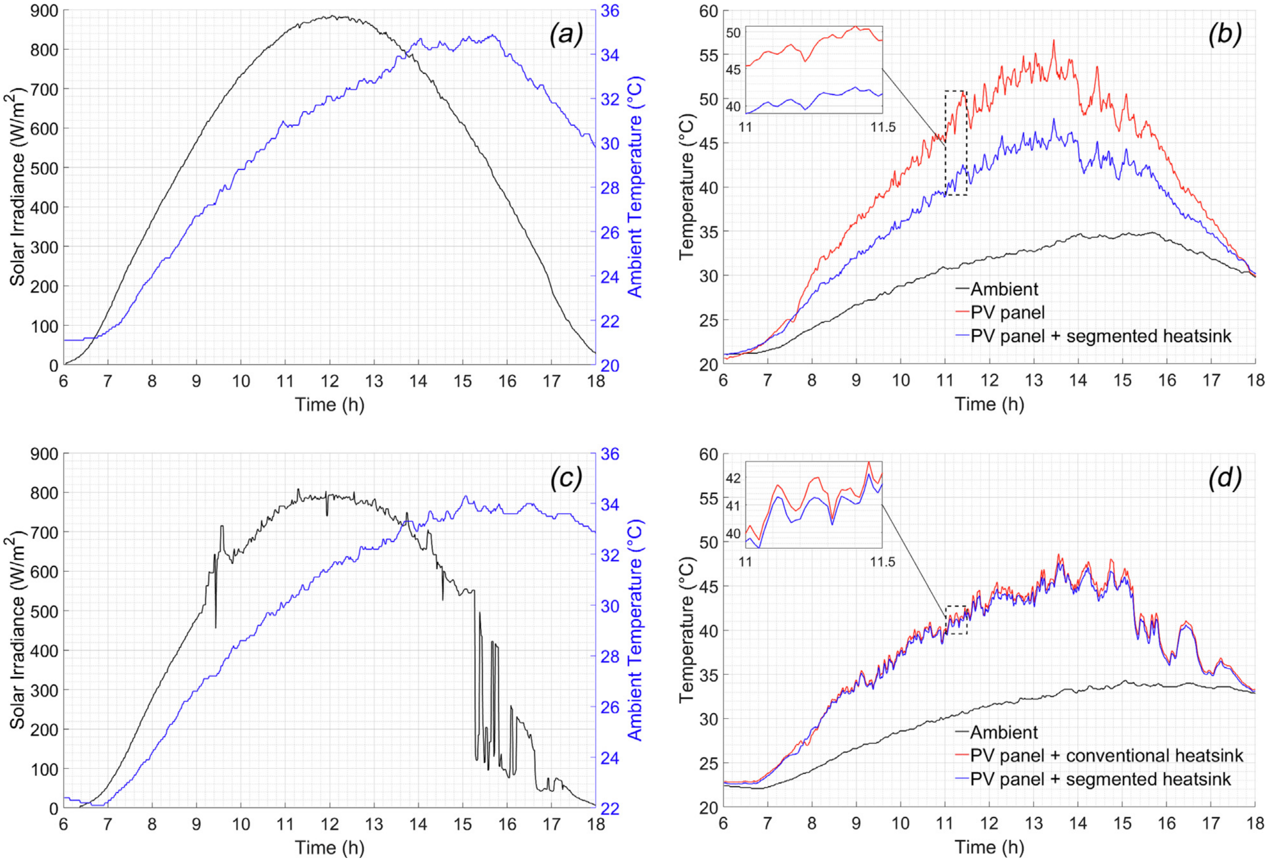

Fig. 9. (a) Irradiance for Test 1, (b) Temperature response of segmented heatsink, Test 1, (c) Irradiance for Test 2, (d) Temperature response between segmented and conventional heatsinks, Test 2.

Fig. 9c shows the irradiance for the comparative test between the heatsinks (Fig. 9d). The difference in the temperature between both systems can reach around 1 °C, in favor of the segmented system, which resembles what was observed at the numerical simulation. However, variable atmospheric conditions such as wind speed and cloudiness reduce effectiveness that could be observed on a typical day. Considering this, the reduction in average temperature for the whole day, for the segmented heatsink when compared to the conventional one, is 0.38 °C. If the average temperature is evaluated during peak hours (high irradiance), where the ambient temperature increases (from 12:00–14:00 h), the reduction would corresponds to 0.54 °C. Due to the necessary orientation to capture the solar radiation, the direction of the wind was unfavorable (coming from the south), given the frontal incidence on the panel front face, so that the interaction of the wind with the heatsink fins was not optimal. However, the amplitude of the thermal cycle, to which the solar panel was exposed, was considerably reduced, and the performance of the conventional heatsink improved without increasing the use of material.

Measurements of the segmented heatsink were conducted over the period of a week in order to compare its energy production with that of the reference PV panel. The electrical performance of the panel was projected using the model of Ishaque et al. [38]. Ambient conditions of test days were varied, including clear, partly cloudy and very cloudy skies, as well as peak irradiance levels ranging from 890 to 1010 W/m^2. During the test week the energy production of the panel increased on average around 2%. Energy yield for a typical day (partly cloudy) for the reference panel was 60.37 Wh. Energy production would have increase 2.35% and 2.2% using the segmented and conventional heatsink, respectively. Throughout testing days, the energy production for the segmented over the conventional heatsink was variable due to atmospheric conditions; however, energy yield for the segmented heatsink was always higher, as the lowest and highest increments observed were 0.134% and 0.35%, respectively. The instantaneous efficiency in hours of peak irradiation increased by around 4%; that is, when the cooling effect of the heatsink is higher, it is possible to reduce the temperature of the PV panel by around 10 °C.

For the construction of the cooling system, 1100 aluminum, gauge 22 (0.71 mm) was chosen because it is a common and economical material (0.5 USD/kg), resistant to corrosion and with good thermal properties. For an array of 1 m^2, a material cost increase of approximately $3.3 could be considered. Another aspect that can impact on the cost, is the weight. PV systems are usually installed in lightweight metal structures that are often mounted on rooftops. Due to this, the increase in the weight of the system is relevant and is a factor that limits the use of many cooling technologies, such as the PV systems with concentration (solar tracking), as they can be deemed complex or expensive. Following the previous example, the heaviest array is also the most effective to reduce the temperature of the unit cell (40 mm fin length). However, if applied in a commercial PV module (60 cells, 19 kg) it would increase the weight by 3 kg/m^2 (26%). This however, could be considered more reasonable and justified if the yield could be obtained with the increased weight of 25–30%.

With this information, using the proposed heatsink, it can be anticipated that for each square meter of PV panel (~250 W), a 94% of PV panel nominal efficiency (90% without fins) could be obtained in a climatic scenario such as the one presented here. This could represent an increase in capacity per square meter of 6 W. If the average cost per installed Watt is considered to be US\$3, an additional US\$18 would be obtained. Therefore, the cost of the material used to improve efficiency (US\$3.3/m^2) could be recovered within 2 years, with the additional benefit of the extended life durability of the panel due to the cooler working conditions.

6. Conclusions

In this study, the performance of different arrangements of a heatsink was tested by numerical simulation. It was found, in a satisfactory way, that the performance of a conventional fin heatsink can be improved in unfavorable weather conditions through a proposed new design, without increasing the use of material. This new design has the potential to be more cost effective, by optimizing the dimensions required to improve the performance of the PV systems that are affected by high temperatures.

A 40 mm length fin was shown to be adequate for this study. It is known that shorter fins can be more efficient, however, with less heat removal capacity. Also, longer fins would increase the heatsink cost without significant contribution. Regarding hydraulic performance, the proposed heatsink model has lower pressure losses compared to a conventional geometry (continuous fin profile) by reducing the velocity losses in the channels, which, added to the geometry of the design, increases the turbulence and significantly favors convective heat transfer.

The maximum temperature reduction observed in the modeling has good agreement to what was observed in the experimental tests (10 °C) during the peak hours of solar irradiance (800–1000 W/m^2). This reduction in temperature means an improvement in the electrical conversion efficiency of the system by ~4% under these conditions. Also, it was observed that the effectiveness of the heatsink increases as the solar irradiation becomes higher (where the delta temperature increases), promoting higher temperature losses. It is considered that by keeping the PV panels working at a lower temperature (around 10 °C under conditions of maximum irradiance), their useful life could be extended by reducing their thermal degradation, especially with solar tracking systems, which are exposed to direct solar radiation for longer periods of time.

In general, the proposed heatsink showed good performance reducing considerably the thermal cycle to which the PV solar panel was exposed, despite the airflow direction not being optimal at this time of the year. Long-term testing would give a detailed performance response from complex natural weather conditions. The development of objective functions (mathematical model design) is also necessary to optimize the design parameters of the heatsink, thus minimizing manufacturing costs, and maximizing efficiency. Determining the optimal values of these parameters is of great importance for the implementation of passive heat dissipation systems in the market.

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